Genre Classification Using the Spotify Tracks Dataset

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1 Project Idea / Design Questions

This project focuses on multi-class music genre classification using structured audio features from the Spotify Tracks Genre dataset. The goal is to evaluate how effectively traditional machine learning models can predict a song's genre based solely on numerical acoustic characteristics.

Design questions include:

- Which machine learning model performs best for multi-class genre classification?
- How do different audio features contribute to genre prediction?
- How do Logistic Regression, Random Forest, and Support Vector Machines compare in terms of accuracy and generalization?

2 Dataset Details

We will use the **Spotify Tracks Genre Dataset** available on Kaggle:

`https://www.kaggle.com/datasets/thedevastator/spotify-tracks-genre-dataset`

The dataset contains approximately 114,000 tracks across 125 distinct genres. Each instance represents a single track and includes numerical audio features extracted from Spotify's API.

Key features include:

- danceability
- energy
- loudness
- speechiness
- acousticness
- instrumentalness
- liveness
- valence
- tempo

The target variable is `track_genre`, making this a supervised multi-class classification problem.

3 Proposed Machine Learning Task

The task is supervised multi-class classification.

Input: Numerical audio features

Output: Predicted genre label

We will implement and compare:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)

4 Software

We will implement the project in Python using Google Colab or a local environment (Jupyter Notebook). Core libraries will include:

- **pandas, numpy** for data loading, cleaning, and numerical operations
- **scikit-learn** for model training and evaluation, preprocessing, and metrics
- **matplotlib** for plots and visualizations
- **imbalanced-learn** for handling class imbalance

We will write code to:

- load and preprocess the dataset (handle missing values, remove duplicates if needed, encode labels)
- split data into training/validation/test sets
- standardize features
- train and tune Logistic Regression, Random Forest, and SVM models
- evaluate models using multiple metrics and generate plots/tables for the report

5 Planned Analysis

Our planned analysis will compare Logistic Regression, Random Forest, and SVM on the same dataset split. We will:

- **Preprocessing:** verify label distribution, handle missing values, and apply feature scaling where appropriate.
- **Baselines:** train a simple baseline for reference.
- **Model training:** train each classifier and use hyperparameter tuning (GridSearchCV or RandomizedSearchCV).
- **Evaluation metrics:** report accuracy, macro-averaged precision/recall/F1, and a confusion matrix for top classes or a selected subset.

- **Generalization:** compare train vs. test performance to check overfitting/underfitting.
- **Interpretability:** analyze feature importance from Random Forest and coefficients from Logistic Regression to identify which audio features most influence predictions.

We will select the best-performing model based on a combination of performance, stability across classes, and reasonable runtime.

6 Breakdown of Work

This project will be completed by a team of 3 students. Responsibilities will be divided as follows:

- **Member 1 - Data and Preprocessing:** Load and clean the dataset, analyze class distribution, handle missing values, encode genre labels, perform feature scaling, and create train/validation/test splits.
- **Member 2 - Model Development:** Implement and tune Logistic Regression and Support Vector Machine models using cross-validation. Compare performance and analyze overfitting/underfitting.
- **Member 3 - Ensemble Model and Evaluation:** Implement and tune the Random Forest model, analyze feature importance, generate confusion matrices, and compute evaluation metrics.

All team members will collaborate on interpreting results, comparing models, preparing visualizations, and writing/editing the final report in Overleaf.

7 Tentative Timeline

- **Phase 1 - Dataset Understanding and Preparation**
  - Download and explore the Spotify Tracks Genre dataset
  - Analyze class distribution and feature statistics
  - Clean data and handle missing values
  - Perform label encoding and feature scaling
  - Create training, validation, and test splits
- **Phase 2 - Baseline Model and Initial Experiments**
  - Train a baseline classifier for comparison
  - Implement Logistic Regression
  - Evaluate initial performance using accuracy and macro F1
- **Phase 3 - Model Development and Tuning**
  - Implement and tune Support Vector Machine
  - Implement and tune Random Forest
  - Perform cross-validation and hyperparameter search
  - Compare training vs. validation performance to check overfitting

- **Phase 4 - Model Comparison and Analysis**
  - Compare all models using consistent evaluation metrics
  - Generate confusion matrices and performance tables
  - Analyze feature importance and model interpretability
- **Phase 5 - Final Evaluation and Report Preparation**
  - Evaluate best model on held-out test set
  - Prepare figures, tables, and experimental results
  - Finalize NeurIPS-format report
  - Clean and document code with README for submission